





Management and Curation of EarthPortal Resources

The O'FAIRe questions in a human-readable format

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1 Introduction

In this short report, our goal is to consider the JSON file that contains the O'FAIRe questions [2] and to convert it into a human-readable format. This should make life easier and the questions better understood, since I will be using them throughout my internship.

2 Development

2.1 The file

The file is called `questions.config.json`. I took it from the source code of O'FAIRe [2], as it was on 4 June 2026. It follows the grid of the O'FAIRe paper [1]: the four FAIR principles are divided into 15 sub-principles, and each sub-principle has its own questions, 61 in total. For every question, the file gives:

- the text of the question;
- the possible answers, each with its points;
- for 38 questions, the list of the metadata properties that the question looks at.

Each sub-principle also carries two numbers. The first, `maxCredits`, is its weight in the final score. The second, `portalMaxCredits`, is described in the documentation of O'FAIRe as the maximum that can be obtained in AgroPortal with the current implementation. Table 1 gives both.

Sub-principle	**Questions**	**With a list of properties**	**Points of the questions**	**maxCredits**	**portalMaxCredits**
F1	4	3	41	41	41
F2	3	3	27	27	27
F3	3	0	42	21	21
F4	3	2	24	24	16
A1	4	3	43	43	43
A1.1	3	0	28	28	28
A1.2	2	0	22	22	22
A2	4	1	20	20	20
I1	5	5	42	44	44
I2	7	5	32	32	22
I3	3	1	33	33	7
R1	6	2	32	32	5
R1.1	3	3	37	37	37
R1.2	8	7	38	38	36
R1.3	3	3	36	36	36
Total	**61**	**38**	**497**	**478**	**405**

Table 1: The 15 sub-principles. *Points of the questions* is the sum of the best answer of each question.

2.2 The 61 questions

The four tables below give the questions, one table per principle. The texts are copied as they are in the file. When the answers of a question only count something, for example the number of properties that are filled, I give them as a small grid instead of repeating the same sentence.

The properties are written with the names used in the file; the three longest lists are in Table 6.

Table 2: Findable: the questions of F1 to F4.

Question	Text	Possible answers, with their points
F1. Ontologies and ontology metadata are assigned a globally unique and persistent identifier. *(maxCredits 41, portalMaxCredits 41)*		
F1Q1	Does the ontology have a "local'' identifier, i.e., a globally unique and potentially permanent identifier assigned by the developer (or developing organization)? Properties: `owl:ontologyIRI`	**0** Ontology URI is absent. **3** Ontology URI is present but invalid. **9** Ontology URI is present and valid.
F1Q2	Does the ontology provide an additional "external'' identifier, i.e., a guarantee globally unique and persistent identifier assigned by an accredited body? If yes, is the external identifier a DOI? Properties: `dct:identifier`	**0** External identifier is absent. **3** External identifier is present but invalid. **6** External identifier is present and valid. **9** External identifier is an invalid DOI. **11** External identifier is a valid DOI.
F1Q3	Are the ontology metadata clearly identified either by the same identifier than the ontology (if included in the ontology file) or with its own globally unique and persistent identifier?	**12** Ontology metadata clearly identified: the repository explicitly identifies its metadata.
F1Q4	Does the ontology provide a version-specific URI, and is this URI resolvable? Properties: `owl:versionIRI`	**0** Version URI is absent. **2** Version URI is present but invalid. **4** Version URI is present, valid but not resolvable. **9** Version URI is present, valid and resolvable.
F2. Ontologies are described with rich ontology metadata. *(maxCredits 27, portalMaxCredits 27)*		
F2Q1	Is the ontology described with additional 'MIRO must' metadata properties? Properties: 24, see Table 6.	*Number of 'MIRO must' properties, and points:* Number 0 1 2 3 4 5 6 7 8+ Points **0 2 4 6 8 10 12 14 16**
F2Q2	Is the ontology described with additional 'MIRO should' or 'optional' metadata properties? Properties: `mod:metrics`, `omv:numberOfClasses`, `omv:numberOfIndividuals`, `omv:numberOfProperties`, `omv:numberOfAxioms`, `vann:preferredNamespacePrefix`	*Number of 'should or optional' properties, and points:* Number 0 1 2 3 4 5 6+ Points **0 1 2 3 3 4 4**
F2Q3	Is the ontology described with another metadata property with no explicit corresponding MIRO requirement? Properties: 37, see Table 6.	*Number of additional properties, and points:* Number 0 1 2 3 4 5 6 7+ Points **0 1 2 3 4 5 6 7**
F3. Ontology metadata clearly and explicitly include the identifier of the ontologies they describe. *(maxCredits 21, portalMaxCredits 21)*		
F3Q1	Are the ontology metadata included and maintained in the ontology file?	**21** *(no text in the file)*
F3Q2	If not, are the ontology metadata described in an external file?	**11** The ontology repository provides ontology metadata in an external file.

Question	Text	Possible answers, with their points
F3Q3	Does that external file explicitly link to the ontology and vice-versa?	**0** No explicit link between the ontology and its metadata and vice-versa. **5** One explicit link between the ontology and its metadata (but no reciprocal). **10** Bidirectional links between the ontology and its metadata.
F4. Ontologies and ontology metadata are registered or indexed in a searchable resource typically an ontology repository. *(maxCredits 24, portalMaxCredits 16)*		
F4Q1	Is the ontology registered in multiple ontology 'libraries'? Properties: `schema:includedInDataCatalog`	*Number of ontology libraries, and points:* Number 0 1 2 3+ Points **0 2 4 6**
F4Q2	Is the ontology registered in multiple open ontology 'repositories'? Properties: `schema:includedInDataCatalog`	*Number of ontology repositories, and points:* Number 0 1 2 3 4 5+ Points **0 2 4 6 8 10**
F4Q3	Are the ontology libraries or repositories properly indexed by Web search engines?	**8** *(no text in the file)*

Table 3: Accessible: the questions of A1 to A2.

Question	Text	Possible answers, with their points
A1. Ontologies and ontology metadata are retrievable by their identifier using a standardized communication protocol. *(maxCredits 43, portalMaxCredits 43)*		
A1Q1	Does the ontology URI and other identifiers, if exist, resolve to the ontology? Properties: `owl:ontologyIRI`, `dct:identifier`	*Number of resolvable identifiers, and points:* Number 0 1 2 Points **0 3 6**
A1Q2	Does the ontology URI (if metadata is included in the ontology file) or the external metadata URI resolve to the metadata record?	**7** The repository provides an external metadata URI which resolves to the metadata record.
A1Q3	Do the ontology URI and the external metadata URI (if the metadata are not included in the ontology file), support content negotiation? Properties: `owl:ontologyIRI`	*Number of negotiation formats, and points:* Number 0 1 2 3 4 5 6 7 8+ Points **0 3 6 9 12 15 18 21 24**
A1Q4	Are the ontology and its metadata accessible through another standard protocol such as SPARQL? Properties: `sd:endpoint`	**0** The ontology is not accessible through a SPARQL endpoint. **6** The ontology declares being accessible through a SPARQL endpoint.
A1.1. The protocol to retrieve ontologies and ontology metadata is open, free and universally implementable. *(maxCredits 28, portalMaxCredits 28)*		
A1.1Q1	Is the ontology relying on HTTP/URIs for its identification and access mechanisms?	**20** The repository uses HTTP/URIs for identification and access mechanisms.
A1.1Q2	Is the ontology access protocol open, free, and universally implementable?	**4** The repository access protocols (HTTP/URIs) are open, free and universally implementable.

Question	Text	Possible answers, with their points
A1.1Q3	If the ontology and metadata are accessible through another protocol, is that protocol open, free, and universally implementable?	**4** The repository uses another protocol (SPARQL) that is open, free and universally implementable.
A1.2. The protocol to retrieve ontologies and ontology metadata supports authentification and authorization when an ontology has access restriction. *(maxCredits 22, portalMaxCredits 22)*		
A1.2Q1	Is the ontology accessible through a protocol that supports authentication and authorization?	**11** The repository supports authentification and authorization when an ontology has access restriction.
A1.2Q2	Are the ontology metadata accessible through a protocol that supports authentification and authorization?	**11** The repository supports authentification and authorization when an ontology has access restriction.
A2. Ontology metadata should be accessible even when the ontology is no longer available. *(maxCredits 20, portalMaxCredits 20)*		
A2Q1	Is the ontology accessible in a repository that supports versioning?	**7** The repository supports versioning.
A2Q2	Are the ontology metadata of each version available?	**5** The repository provides metadata for each version of the ontology.
A2Q3	Are the ontology metadata accessible even if no more versions of the ontology are available?	**4** The repository provides metadata even if no more versions of the ontology are available.
A2Q4	Is the status of the ontology clearly informed? Properties: `omv:status`, `owl:deprecated`	**0** Ontology status absent. **2** Ontology status partially specified. **4** Ontology status fully specified.

Table 4: Interoperable: the questions of I1 to I3.

Question	Text	Possible answers, with their points
I1. Ontologies and ontology metadata use a formal, accessible, shared and broadly applicable language for knowledge representation. *(maxCredits 44, portalMaxCredits 44)*		
I1Q1	What is the representation language used for ontology and ontology metadata? Properties: `omv:hasOntologyLanguage`	**0** Ontology knowledge representation language absent. **5** Ontology and ontology metadata are represented in PDF. **7** Ontology and ontology metadata are represented in TXT. **9** Ontology and ontology metadata are represented in CSV. **10** Ontology and ontology metadata are represented in XML. **11** Ontology and ontology metadata are represented in OBO. **12** Ontology and ontology metadata are represented in RDFS. **15** Ontology and ontology metadata are represented in SKOS. **18** Ontology and ontology metadata are represented in OWL.
I1Q2	Is the representation language used a W3C Recommendation? Properties: `omv:hasOntologyLanguage`	**0** Ontology and ontology metadata are not represented using a W3C Recommendation. **10** Ontology and ontology metadata are represented using a W3C Recommendatione.

Question	Text	Possible answers, with their points
I1Q3	Is the syntax of the ontology informed? Properties: `omv:hasOntologySyntax`	**0** Ontology syntax is absent. **5** Ontology syntax is present.
I1Q4	Is the formality level of the ontology informed? Properties: `omv:hasFormalityLevel`	**0** Ontology formality level absent. **5** Ontology formality level present.
I1Q5	Is the availability of other syntaxes/formats informed? Properties: `dct:hasFormat`, `dct:isFormatOf`	**0** No information is provided about the availability of the ontology in other formats. **4** The availability of other ontology formats is informed.
I2. Ontologies and ontology metadata use vocabularies that follow FAIR principles. *(maxCredits 32, portalMaxCredits 22)*		
I2Q1	Does the ontology import other FAIR vocabularies? Properties: `owl:imports`	**0** The ontology does not import other vocabularies. **5** The ontology imports other vocabularies (independently of their level of FAIRness).
I2Q2	Does the ontology reuse terms from other FAIR vocabularies (URIs)? Properties: `dct:relation`	**0** The ontology does not declare reusing other vocabularies. **5** The ontology declares reusing terms from other vocabularies (independently of their level of FAIRness).
I2Q3	If yes, does it include the minimum information for those terms?	**3** *(no text in the file)*
I2Q4	Is the ontology aligned to other FAIR vocabularies? Properties: `voaf:hasEquivalencesWith`	**0** The ontology does not declare being aligned to other vocabularies. **5** The ontology declares being aligned to other vocabularies (independently of their level of FAIRness).
I2Q5	If yes, are those alignments well represented and to unambiguous entities? If yes, are those alignements curated?	**7** *(no text in the file)*
I2Q6	Does the ontology provide metadata information about relation to or influence of other FAIR vocabularies? Properties: `voaf:specializes`, `schema:translationOfWork`, `voaf:similar`, `voaf:generalizes`, `dct:isPartOf`	**0** The ontology does not declare influential relations to other vocabularies. **2** The ontology declares influential relations to other vocabularies (independently of their level of FAIRness).
I2Q7	Does the ontology reuse standard and FAIR metadata vocabularies to describe its metadata? Properties: `voaf:metadataVoc`	**0** The ontology does not declare reusing standards metadata vocabularies to describe its metadata. **5** The ontology declares reusing standards metadata vocabularies (more or less 'FAIR') to describe its metadata.
I3. Ontologies or ontology metadata include qualified references to other (meta)data. *(maxCredits 33, portalMaxCredits 7)*		
I3Q1	Does an ontology provide qualified cross-references to external resources/databases?	**20** *(no text in the file)*
I3Q2	If yes, are those cross-references well represented and to unambiguous entities?	**6** *(no text in the file)*

I3Q3	Does the ontology use valid URIs to encode some metadata values? Properties: 30, see Table 6.	*Number of metadata values encoded as URIs, and points:* Number 0 1 2 3 4 5 6 7+ Points **0 1 2 3 4 5 6 7**

Table 5: Reusable: the questions of R1 to R1.3.

Question	Text	Possible answers, with their points
R1. Ontologies and ontology metadata are richly described with a plurality of accurate and relevant attributes. *(maxCredits 32, portalMaxCredits 5)*		
R1Q1	Does the ontology provide information about how classes or concepts are defined? Properties: `mod:prefLabelProperty`, `mod:synonymProperty`, `mod:definitionProperty`, `mod:authorProperty`, `mod:obsoleteProperty`	*Number of properties that inform how classes or concepts are defined, and points:* Number 0 1 2 3 4 5 Points **0 1 2 3 4 5**
R1Q2	Does the ontology provide information about its hierarchy? Properties: `mod:hierarchyProperty`, `mod:obsoleteParent`, `mod:maxDepth`	*Number of properties that inform about the hierarchy, and points:* Number 0 1 2 3 Points **0 1 2 3**
R1Q3	How much of the ontology objects are described with labels?	**7** *(no text in the file)*
R1Q4	How much of the ontology objects are defined using a text description?	**6** *(no text in the file)*
R1Q5	How much ontology objects are defined using a property restriction or an equivalent class?	**6** *(no text in the file)*
R1Q6	How much of the ontology objects provide provenance information with annotation properties?	**5** *(no text in the file)*
R1.1. Ontologies and ontology metadata are released with a clear and accessible usage license. *(maxCredits 37, portalMaxCredits 37)*		
R1.1Q1	Is the ontology license clearly specified, with an URI that is resolvable and supports content negotiation? Properties: `dct:license`	**0** Ontology license absent. **4** Ontology license present but not a valid URI. **8** Ontology license present and valid but not resolvable. **12** Ontology license present, valid, resolvable but with no content negotiation. **15** Ontology license is clearly specified with a valid and resolvable URI that supports content negotiation.
R1.1Q2	Are the ontology access rights specified and permissions documented? Properties: `dct:accessRights`, `cc:morePermissions`	**0** Ontology access rights and permissions absent. **4** Ontology access rights are clearly specified. **7** Ontology access rights are clearly specified and permissions are documented.
R1.1Q3	Are the ontology usage guidelines and copyright holder documented? Properties: `cc:useGuidelines`, `dct:rightsHolder`	**0** Ontology usage guidelines and copyright holder are absent. **7** Ontology usage guidelines or copyright holder is absent. **15** Ontology usage guidelines and copyright holder are present.

Question	Text	Possible answers, with their points
R1.2. Ontologies and ontology metadata are associated with detailed provenance. *(maxCredits 38, portalMaxCredits 36)*		
R1.2Q1	Does the ontology provide information about the actors involved in its development? Properties: `dct:creator`, `pav:curatedBy`, `dct:contributor`, `schema:translator`	*Number of properties about the actors involved in the development, and points:* Number 0 1 2 3 4 Points **0 3 5 7 8**
R1.2Q2	Does an ontology provide information on its general provenance? Properties: `dct:source`, `prov:wasGeneratedBy`, `prov:wasInvalidatedBy`	*Number of properties about the general provenance, and points:* Number 0 1 2 3 Points **0 2 4 6**
R1.2Q3	Are the accrual methods and policy of the ontology documented? Properties: `dct:accrualMethod`, `dct:accrualPeriodicity`, `dct:accrualPolicy`	*Number of accrual properties, and points:* Number 0 1 2 3 Points **0 2 4 6**
R1.2Q4	Is the ontology clearly versioned with version information and links to previous versions? Properties: `owl:versionInfo`, `owl:priorVersion`, `dct:hasVersion`	*Number of properties about versioning, and points:* Number 0 1 2+ Points **0 2 4**
R1.2Q5	Are the ontology latest changes documented?	**2** *(no text in the file)*
R1.2Q6	Are the methodology and tools used to build the ontology documented? Properties: `omv:usedOntologyEngineeringTool`, `omv:usedOntologyEngineeringMethodology`, `omv:conformsToKnowledgeRepresentationParadigm`	*Number of properties about the methodology and tools used, and points:* Number 0 1 2 3 Points **0 2 4 6**
R1.2Q7	Is the ontology rationale documented? Properties: `omv:designedForOntologyTask`, `mod:competencyQuestion`	*Number of properties about the rationale, and points:* Number 0 1 2 Points **0 2 4**
R1.2Q8	Does an ontology inform about its funding organization? Properties: `foaf:fundedBy`	**0** The ontology does not inform about its funding organization. **2** The ontology informs about its funding organization.
R1.3. Ontologies and ontology metadata meet domain-relevant community standards. *(maxCredits 36, portalMaxCredits 36)*		
R1.3Q1	Does an ontology provide information about projects using or organizations endorsing? Properties: `mod:ontologyInUse`, `omv:endorsedBy`	**0** The ontology does not provide information about projects or endorsing organizations. **5** The ontology provide information about projects or endorsing organizations. **10** The ontology provide information about projects and endorsing organizations.

Question	Text	Possible answers, with their points
R1.3Q2	Is the ontology included in a specific community set or group? Properties: `mod:group`	**0** The ontology does not provide information about its community set or group. **10** The ontology is included in one of this groups: "WHEAT", "CROP", "INRAE". **15** The ontology is included in the OBO library. **20** The ontology is included in the OBO library and is part of the OBO Foundry.
R1.3Q3	Is the ontology openly and freely available? Properties: `dct:accessRights`	**0** The ontology is not openly and freely available. **6** The ontology is openly and freely available.

Table 6: The three long lists of properties, in the order of the file.

Question	N	Properties
F2Q1	24	`omv:acronym`, `dct:title`, `dct:alternative`, `skos:hiddenLabel`, `dct:description`, `foaf:page`, `omv:resourceLocator`, `omv:keywords`, `dct:coverage`, `foaf:homepage`, `vann:example`, `vann:preferredNamespaceUri`, `void:uriRegexPattern`, `idot:exampleIdentifier`, `dct:publisher`, `dct:subject`, `owl:backwardCompatibleWith`, `door:comesFromTheSameDomain`, `omv:knownUsage`, `dct:audience`, `doap:repository`, `doap:bugDatabase`, `doap:mailing-list`, `mod:hasEvaluation`
F2Q3	37	`dct:language`, `dct:abstract`, `mod:analytics`, `dct:bibliographicCitation`, `rdfs:comment`, `foaf:depiction`, `foaf:logo`, `voaf:toDoList`, `schema:award`, `schema:associatedMedia`, `owl:incompatibleWith`, `dct:hasPart`, `schema:workTranslation`, `door:hasDisparateModelling`, `voaf:usedBy`, `voaf:hasDisjunctionsWith`, `omv:keyClasses`, `void:rootResource`, `mod:browsingUI`, `void:propertyPartition`, `void;classPartition`, `void:dataDump`, `void:openSearchDescription`, `void:uriLookupEndpoint`, `schema:comment`, `dct:created`, `dct:modified`, `dct:valid`, `dct:dateSubmitted`, `pav:curatedOn`, `omv:isOfType`, `void:propertyPartition`, `void:classPartition`, `mod:classesWithMoreThan25Children`, `mod:classesWithOneChild`, `mod:classesWithNoDefinition`, `mod:maxChildCount`
I3Q3	30	`owl:imports`, `owl:priorVersion`, `owl:backwardCompatibleWith`, `owl:incompatibleWith`, `dct:isPartOf`, `dct:hasPart`, `dct:hasVersion`, `dct:isFormatOf`, `dct:hasFormat`, `dct:relation`, `voaf:specializes`, `schema:workTranslation`, `schema:translationOfWork`, `door:comesFromTheSameDomain`, `voaf:similar`, `voaf:hasEquivalencesWith`, `door:hasDisparateModelling`, `voaf:usedBy`, `voaf:generalizes`, `voaf:hasDisjunctionsWith`, `omv:hasFormalityLevel`, `dct:license`, `omv:hasOntologySyntax`, `dct:subject`, `mod:prefLabelProperty`, `mod:definitionProperty`, `mod:synonymProperty`, `mod:authorProperty`, `mod:hierarchyProperty`, `mod:obsoleteProperty`

3 Conclusion

While converting the file, I noticed a few things that are worth mentioning:

- The points of the questions add up to 497, while the `maxCredits` add up to 478. The difference comes from two sub-principles: the questions of F3 add up to 42 for a `maxCredits` of 21, and those of I1 add up to 42 for a `maxCredits` of 44.
- 23 questions have no list of properties.
- The list of F2Q3 contains `void:propertyPartition` twice, and `void;classPartition` written once with a semicolon.

References

[1] Emna Amdouni, Syphax Bouazzouni, and Clement Jonquet. O'FAIRe makes you an offer: metadata-based automatic FAIRness assessment for ontologies and semantic resources. *International Journal of Metadata, Semantics and Ontologies*, 16(1):16–46, 2022.

[2] O'FAIRe configuration files: metadata properties and assessment questions. `https://github.com/agroportal/fairness/tree/master/src/main/resources/config/common`. Files `properties.config.json` and `questions.config.json`, at commit `24def6b` of 4 June 2026.